Weilue Luo

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Education

University College London Computer Vision (Distinction - avg. 85%) *London, UK* *Sep. 2021 - Sep. 2022*

- **Relevant Modules** Networked Systems, Machine Learning for Visual Computing, Image Processing, Machine Vision, Computer Graphics, Perception and Interface, Robot Vision and Navigation, and Virtual Environment.

University of Manchester Artificial Intelligence (Distinction - avg. 83%) *Manchester, UK* *Sep. 2018 - Sep. 2021*

- **Relevant Modules** Mathematics, Operating System, Algorithms and Complexity, Databases, Logic and Modelling, Software Engineering, Imperative Programming, Object-oriented Programming, Machine Learning and Optimization, AI and Games, Computer Vision, Graphics and Image Processing, Distributed Computing, Mobile System, and Cryptography.

Bellerbys College Computing (1st in cohort - avg. 84%) *Brighton, UK* *Jan. 2018 - Jun. 2018*

- **Relevant Modules** Decision & Pure Maths, Databases, Programming & Code, Web Design, and System Analysis.

Experience

Microsoft Software Engineer IC3 - C# *Nov. 2024 - Current*

- Azure Fabric Data team. Developing data streaming systems behind PowerBi which serves millions of customers worldwide.
- Built, integrated, and presented an AI Agent development workflow to teams.

Cisco Software Engineer - Java & Go & Typescript *Oct. 2022 - Sep. 2024*

- Integrated external services and applications in cloud environments (AWS and Azure), reducing manual steps by up to 60%.
- Worked with teams to move authentication and authorization services from self-hosted to cloud-managed databases, ensuring a smooth transition.
- Led the development of AWS Terraform plugins for Cisco Cloud, gaining hundreds of users in the initial release.
- Designed and implemented external platforms, developing key public APIs, attracting hundreds of thousands of requests during beta release. Link: (Cisco Defense Orchestrator Documentation).
- Developed an easy-to-use internal rate limit library, used by several internal applications.

Amazon Software Engineer Internship - Java *Jun. 2021 - Sep. 2021*

- Built and embedded an experiment framework into the existing question-answering pipeline. Conducted experiments with 1% customer traffic, and presented a highly-rated proof-of-concept report.
- Developed an efficient workflow for future experiments, and reduced experiment deployment time needed from days to hours.
- Received full-time offer (Organization wide layoff).

Amazon Software Engineer Internship - Java *Jun. 2020 - Sep. 2020*

- Developed procedures that aggregate ≈3 billion item records into concise datasets daily using tools based on Spark.
- Reduced time-consuming ad-hoc graph analysis from minutes to seconds by developing an effective modern visualization dashboard using Elastic Search and Kibana.
- Developed an API package for automating the creation of visualization components and dashboards.
- Received full-time offer.

Events

Win-a-Mac Hack (Agentic Edition) 2nd Place *Mar. 2026*

- Encode Club. Built “AI Storyboard Dungeon”, a multi-agent system where AI personas with distinct personalities debate and collaboratively generate visual storyboards in a real-time 3D environment.

Cambridge AI RealityX Hackathon 2026 1st Place *Mar. 2026*

- Medical Healthcare Track. Built “Mind The Gap”, an AR-based post-stroke rehabilitation coach using pose detection and 3D body models (SMPL-X) to guide patients through recovery exercises at home.

Agentic AI Hackathon 2nd Place *Dec. 2025*

- SpoonOS and Neo challenge. Built AI an agent called CoSign.ai acting as a service layer to verify payments are authentic and secure.

London AI Hackathon 2025 2nd Place *Feb. 2025*

- Encode Club. Created tutorials on deploying and building with SpoonOS with AI.

University of Manchester Software Engineer Mentor *Oct. 2022 - Current*

- Mentored a group of 8 second year undergraduate students as they contribute to a open source project on a yearly basis.
- Received positive feedback from students and faculty members.

NoGhost Data Researcher - Python *Jun. 2022 - Sep. 2022*

- Built multi-processing data pipeline, improved and implemented state-of-the-art machine learning papers.
- Part of Masterś degree project at UCL, condensed a thesis received distinction grade, rated by engineers and professors from UCL.

Selected Projects

Aex – Agent Execution Platform & SDK TypeScript, Next.js, Node.js, Supabase, Postgres, Vercel, Railway, Anthropic Claude *2025 - Present*

- Platform plus published `@aexhq/sdk` and `@aexhq/cli` npm packages for running autonomous agents with durable run lifecycle, output capture, and policy-driven cleanup. Monorepo of Next.js dashboard (BFF/UI), long-running worker, shared DB/Vault/Storage adapters, and end-to-end live test gates. Link: https://aex.dev, https://github.com/aexhq/aex.

Brikell – AI Due-Diligence Platform for Danish Properties Next.js, TypeScript, Supabase (Postgres + Storage + RLS), Anthropic Claude, MCP, Vercel *2025 - Present*

- Full-stack AI-native platform automating property due diligence via a Claude agent orchestrated over MCP datasource providers (Datafordeler, Dataforsyningen, Plandata). Durable report lifecycle, owner-scoped Vault search across reports and source-document evidence, and a grounded read-only assistant. Link: https://brikell.dk.

B-Roll – Historical Footage Search Engine for Documentaries Python, FastAPI, Next.js, LanceDB (hybrid BM25 + dense vector), Gemini 2.5 Flash, MCP *2025 - Present*

- Search "flywheel" that closes the semantic gap between documentary narration and visual B-roll. Parses subtitles into 15s windows, uses an LLM "director" to produce visual queries, runs hybrid local search (Tantivy BM25 + text-embedding-004) alongside cloud retrieval, and ingests VLM-generated 500-word clip descriptions back into LanceDB so the local index gets smarter over time. Multi-agent collection (Agent1) routes through MCP providers and writes compact source records for context-first reranking (Agent2).

AI-Powered Business Websites Next.js, React, TypeScript, Tailwind CSS, Three.js, Framer Motion, Vercel *2025 - Present*

- Utilized AI to build production websites for local companies, including a MedTech startup showcasing 3D needle-tip tracking technology for ultrasound-guided procedures, and a childcare nursery with contact forms, responsive design, and email integration.
- Applied SEO and Generative Engine Optimization (GEO) — structured data, semantic markup, and LLM-friendly content surfaces — so client sites are discoverable in both traditional search and AI-assistant answers.

CellNext – 3D Jewelry Designer Next.js, TypeScript, Rust, WebAssembly, WebGL *2025 - Present*

- A web-based 3D cellular jewelry mesh designer with a custom Rust/WASM engine for topology generation, physics simulation, 2D→3D transform, mesh generation, and Catmull-Clark subdivision. Link: https://cells.lossgrad.com/.

PBM – AI Property Compliance Platform Next.js, TypeScript, Supabase, OpenAI, GoCardless, Vercel *2025 - Present*

- Full-stack AI-native SaaS for property compliance management. Features include AI-powered document QA chat, automated compliance task generation with scoring, property/tenancy document tracking, OAuth authentication, and payment processing.

Nvwa – AI Newsletter Aggregation Python, Next.js, Azure OpenAI (GPT-4o), Azure Cosmos DB, HDBSCAN *2025 - Present*

- AI-powered content aggregation and email digest platform. Multi-stage filtering pipeline (linguistic heuristics, sentiment analysis, semantic density, LLM validation), semantic clustering via HDBSCAN, and GPT-4o summarization. Scrapes Reddit, HackerNews, and X/Twitter. Deployed on Azure Container Apps.

Collaborate Live HTML/CSS, and Javascript *Jan. 2019 - Mar. 2019*

- Led 7 students to build a web application for a code-based interview. Features include real-time coding, drawing, compliation and terminal.
- Awarded 1st place in most appealing idea and highest technical quality prize (2/4 prizes) out of 37 teams.

Local Nameserver Python *Feb. 2022 - Mar. 2022*

- Written a local nameserver that able to resolve any IPv4 address according to RFC 1034 and RFC 1035 specification.

Poisson Image Editing Sole Developer - Python *Dec. 2021 - Jan. 2022*

- Implemented the Poisson Image Editing paper. Including techniques such as naive filling; import & mixing gradient; texture flattening, colour isolation & illumination. The project can be found at *https://github.com/Redcxx/poisson-image-editing*.

Skills

Programming	(>10k lines) Java,Go,Typescript/Javascript,Python,HTML/CSS;(>2k lines) Matlab,C,GLSL.
Tech Stack	Spring,GraphQL,OpenAPI,NodeJs,Jenkins,Angular, ReactJs,ThreeJs,NextJs,SASS/SCSS,.
Environments	Git,CR,CI/CD,AWS,Terraform,Scrum,Agile,Markdown,Linux/Windows,LaTeX.
Language	(Native) Chinese and Cantonese;(Fluent) English;(New) Japanese